Analysis III
Home Work 4

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October 18, 2026

Exercise 1 (Fubini's Theorem, 2+3+1 P.) Define $f : [0;1]^2 \to \mathbb{R}$ by $f(0,0) = 0$ and

$$f(x,y) = \frac{y^2 - x^2}{(x^2+y^2)^2}$$

for $(x,y) \neq (0,0)$.

(a) Prove that, for $(x,y) \in (0;1)^2$,

$$f(x,y) = \frac{\partial}{\partial x}\frac{x}{x^2+y^2} = -\frac{\partial}{\partial y}\frac{y}{x^2+y^2}.$$

(b) Compute the integrals

$$\int_0^1 \left[\int_0^1 f(x,y)\,dx\right] dy \qquad \text{and} \qquad \int_0^1 \left[\int_0^1 f(x,y)\,dy\right] dx.$$

(c) Is $f \in L^1([0;1]^2;\mathbb{R})$? Justify.

Answer:

(a) We have that:

$$\begin{aligned}\frac{\partial}{\partial x} x(x^2+y^2)^{-1} &= (x^2+y^2)^{-1} + x\left[\frac{-1}{(x^2+y^2)^2}\cdot 2x\right] \\ &= \frac{1}{(x^2+y^2)^2}\left[x^2+y^2-2x^2\right] \\ &= \frac{y^2-x^2}{(x^2+y^2)^2}.\end{aligned}$$

Similarly:

$$\frac{\partial}{\partial y} y(x^2+y^2)^{-1} = \frac{x^2-y^2}{(x^2+y^2)^2}.$$

i.e.

$$\begin{aligned}\frac{\partial}{\partial x} x(x^2+y^2)^{-1} &= -\frac{\partial}{\partial y} y(x^2+y^2)^{-1} \\ &= f(x,y)\end{aligned}$$

(b) For each fixed $y > 0$, the function $x \mapsto f(x,y)$ is continuous on $[0;1]$. Its Lebesgue integral agrees with its Riemann integral by Theorem 11.2.17(1). Using the antiderivative from (a), we obtain

$$\int_0^1 f(x,y)\,dx = \left[\frac{x}{x^2+y^2}\right]_{x=0}^{x=1} = \frac{1}{1+y^2}.$$

Similarly, for each fixed $x > 0$, the function $y \mapsto f(x, y)$ is continuous on $[0; 1]$, so Theorem 11.2.17(1) and part (a) give

$$\int_0^1 f(x, y)\, dy = -\left[\frac{y}{x^2 + y^2}\right]_{y=0}^{y=1} = -\frac{1}{1 + x^2}.$$

The exceptional parameter values $y = 0$ and $x = 0$ form sets of measure zero by Example 11.1.4(1). We may define the inner-integral functions arbitrarily at these exceptional values: Theorem 11.2.15 ensures that this does not affect the outer integrals. Applying Theorem 11.2.17(1) to the continuous functions $t \mapsto \pm(1 + t^2)^{-1}$ gives

$$\int_0^1 \left[\int_0^1 f(x, y)\, dx\right] dy = \int_0^1 \frac{1}{1 + y^2}\, dy = [\arctan y]_0^1 = \frac{\pi}{4},$$
$$\int_0^1 \left[\int_0^1 f(x, y)\, dy\right] dx = -\int_0^1 \frac{1}{1 + x^2}\, dx = -\frac{\pi}{4}.$$

(c) If $f \in L^1([0; 1]^2; \mathbb{R})$, Fubini's Theorem (Theorem 11.3.3), applied to the extension of f by zero outside $[0; 1]^2$, would imply that the two iterated integrals in (b) are equal. Since $\pi/4 \neq -\pi/4$, we conclude that

$$f \notin L^1([0; 1]^2; \mathbb{R}).$$

Exercise 2 (Application of the MCT, 2+2 P.)

Let $f \in L^1(\mathbb{R}^n;\mathbb{R})$ and denote by $B_\varepsilon(x)$ the ball of radius $\varepsilon > 0$ around $x \in \mathbb{R}^n$. We want to prove that

$$\lim_{\varepsilon \to 0^+} \int_{B_\varepsilon(x)} f(y)dy = 0.$$

(a) Use the Monotone Convergence Theorem to prove the assertion under the additional assumption that $f \geq 0$.

Hint: You may use without proof that $f\chi_B \in L^1(\mathbb{R}^n;\mathbb{R})$ for any ball $B \subset \mathbb{R}^n$. You may also assume that ε decreases to 0 monotonically.

(b) Prove the assertion without the additional assumption by reducing the general problem to the situation in (a).

Hint: Sheet 3, Exercise 1 might be useful.

Remark: It is in general a good strategy to prove statements first for non-negative functions and then generalize. Answer:

(a) Fix $x \in \mathbb{R}^n$ and assume $f \geq 0$. For $k \in \mathbb{N}$, $k \geq 1$, define

$$h_k(y) = f(y)\big(1 - \chi_{B_{1/k}(x)}(y)\big).$$

By the hint, $f\chi_{B_{1/k}(x)} \in L^1$, so $h_k \in L^1$ by the linearity statement in Theorem 11.2.14(1). Since the balls decrease as k increases,

$$0 \leq h_k \leq h_{k+1} \leq f.$$

For every $y \neq x$, eventually $1/k < \|y - x\|$, and hence $h_k(y) = f(y)$. Thus $h_k \to f$ pointwise almost everywhere, since $\{x\}$ has measure zero by Example 11.1.4(1). By Theorem 11.2.14(1)–(2),

$$0 \leq \int_{\mathbb{R}^n} h_k(y)\,dy \leq \int_{\mathbb{R}^n} f(y)\,dy < \infty.$$

Thus the sequence of integrals is bounded, and the Monotone Convergence Theorem (Theorem 11.3.7) gives

$$\int_{\mathbb{R}^n} h_k(y)\,dy \longrightarrow \int_{\mathbb{R}^n} f(y)\,dy.$$

By Theorem 11.2.14(1) (linearity),

$$\int_{B_{1/k}(x)} f(y)\,dy = \int_{\mathbb{R}^n} f(y)\,dy - \int_{\mathbb{R}^n} h_k(y)\,dy \longrightarrow 0.$$

To obtain the limit for arbitrary $\varepsilon \to 0^+$, Theorem 11.2.14(1)–(2) gives, for $0 < \varepsilon \le 1/k$,

$$0 \le \int_{B_\varepsilon(x)} f(y)\,dy \le \int_{B_{1/k}(x)} f(y)\,dy.$$

The right-hand side tends to zero, proving the assertion.

(b) For general $f \in L^1(\mathbb{R}^n;\mathbb{R})$, write

$$\begin{aligned} f &= f^+ - f^-, \\ f^+ &= \max\{f, 0\}, \qquad f^- = \max\{-f, 0\}. \end{aligned}$$

By Theorem 11.2.14(3), $|f| \in L^1$. Since

$$f^+ = \frac{|f| + f}{2}, \qquad f^- = \frac{|f| - f}{2},$$

Theorem 11.2.14(1) implies $f^+, f^- \in L^1$. Both are non-negative. Applying (a) to each and using Theorem 11.2.14(1) gives

$$\begin{aligned} \lim_{\varepsilon\to 0^+} \int_{B_\varepsilon(x)} f(y)\,dy &= \lim_{\varepsilon\to 0^+} \int_{B_\varepsilon(x)} f^+(y)\,dy - \lim_{\varepsilon\to 0^+} \int_{B_\varepsilon(x)} f^-(y)\,dy \\ &= 0 - 0 = 0. \end{aligned}$$